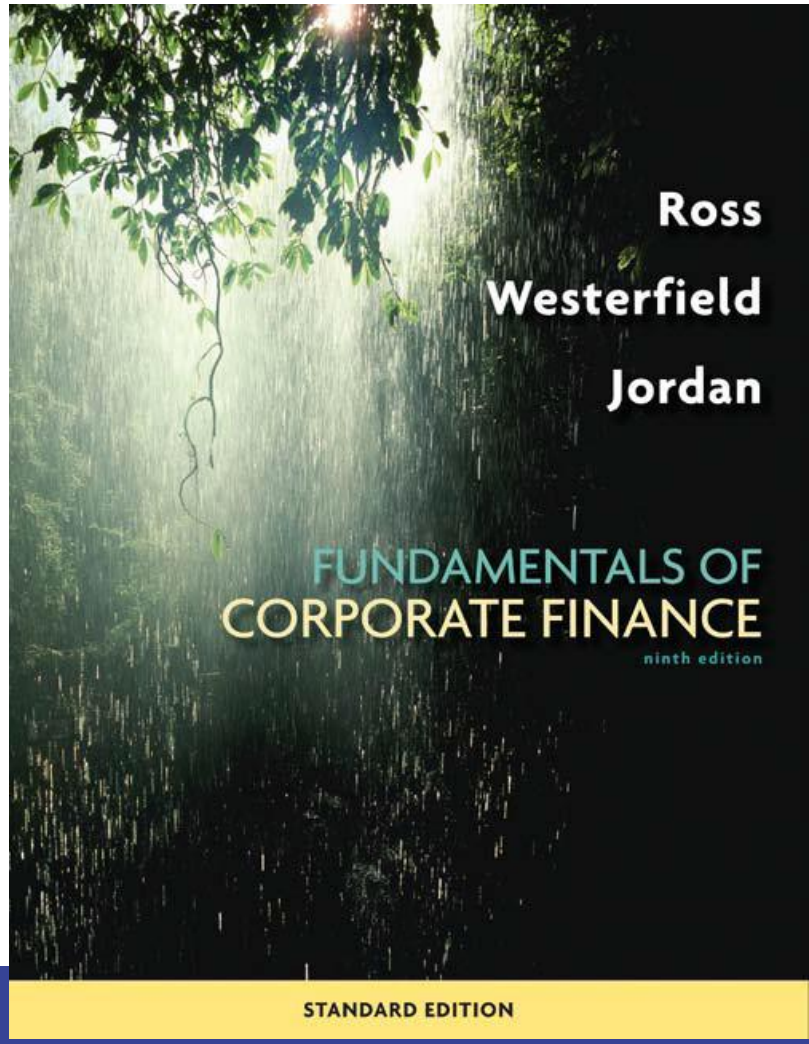
Fundamental of Corporate Finance

Standard Edition

Ross, Westerfield, Jordan

Lecturer:
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Chapter 13: Leverage and Capital Structure



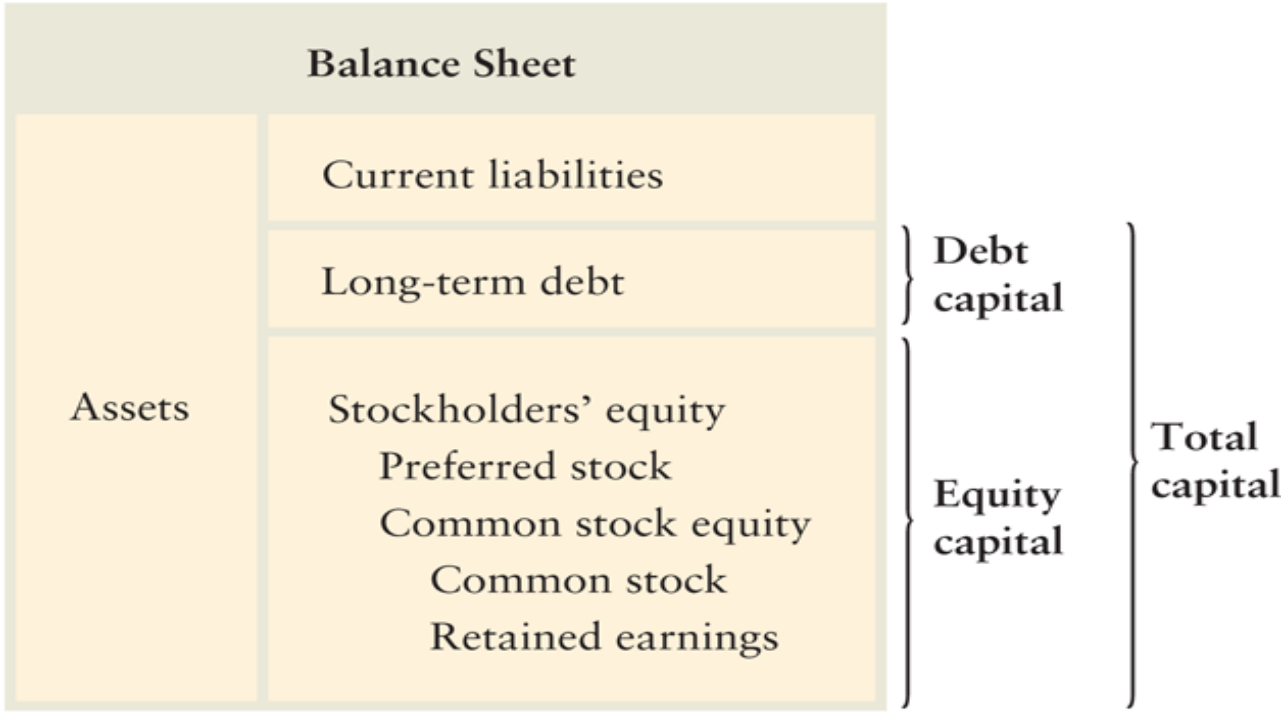
Key Formulas

Concept	Formula
Debt-to-Equity Ratio	D/E=Total Debt/Total Equity
M&M Proposition I (No Taxes)	VL=VU *VU=EBIT/ Keu*
M&M Proposition II (No Taxes)	Ke=Keu+(Keu-Kd)×(D/E)
M&M Proposition I (With Taxes)	VL=VU+Tc×D
M&M Proposition II (With Taxes)	Ke=Keu+(1-Tc)(Keu-Kd)(D/E)
Trade-off Theory (Value)	VL=VU+TcD-PV(Distress Costs)
WACC	WACC=(E/V)Ke+[(D/V)Kd(1-Tc)]
EBIT-EPS Indifference Point	(EBIT-I1)(1-T)/E1=(EBIT-I2)(1-T)/E2
Zero Growth Model	P= EPS/ r
Interest	I=D×Kd
Equity earnings	Equity earnings: EBIT-I

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The Firm's Capital Structure: Types of Capital

All of the items on the right-hand side of the firm's balance sheet, excluding current liabilities, are sources of capital. The following simplified balance sheet illustrates the basic breakdown of total capital into its two components, debt capital and equity capital.



Capital Structure Theory

- Includes:
- A. Modigliani and Miller (M&M) Propositions
- B. Trade off theory

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A. Modigliani and Miller (M&M) Propositions

M&M Proposition I (No Taxes, 1958):

$V_L = V_U$

- V_L is the value of a levered firm
- V_U is the value of an unlevered firm

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Example

- Two firms, U and L, have identical assets and business risk
- Expected EBIT = $1,000,000 per year forever
- Unlevered cost of equity, Keu=10%
- Firm L has $4,000,000 perpetual debt at Kd=5%, while Firm U is all-equity financed. Calculate

1- value of unlevered company: $VU=EBIT/Keu$ =1,000,000/0.10=$10,000,000

2- Value of Equity: Since Firm L has debt of $4,000,000:

E=VL-D=10,000,000-4,000,000=$6,000,000

3- for L: interest- equity earning- cost of equity

Interest expense: I=D×Kd=4,000,000×5%=$200,000

Equity earnings: EBIT-I=1,000,000-200,000=$800,000

Cost of equity: Ke=800,000/6,000,000=13.333%

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M&M Propositions II (No Taxes)

The cost of equity increases with leverage because shareholders demand a higher return for the increased financial risk.

$$K_e = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

- *Ke* is the cost of equity of a levered firm
- *Keu* is the cost of equity of an unlevered firm
- *Kd* is the cost of debt
- *D*/*E* is the debt-to-equity ratio.

The WACC remains constant.

Using **M&M Proposition II (No Taxes)**:

$$K_e = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

Based on the above information:

Ke=10%+(10%-5%)(4,000,000/6,000,000)=13.333%

WACC

$$\text{WACC} = \left(\frac{E}{V} \times K_e\right) + \left(\frac{D}{V} \times K_d \times (1 - T_c)\right)$$

WACC=(6/10)(13.333%)+(4/10)(5%)=8%+2%=10%

M&M with Corporate Taxes (1963) Proposition I (with taxes):

When taxes are introduced, debt becomes valuable because interest payments are tax-deductible.

$$V_L = V_U + T_c \times D$$

- Tc is the corporate tax rate
- $Tc \times D$ is the present value of the interest tax shield.

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Proposition II (with taxes):

$$K_e = K_{eu} + (1 - T_c) \times (K_{eu} - K_d) \times \frac{D}{E}$$

- The WACC declines as debt increases because the tax shield lowers the effective cost of debt

Firm A is currently an all-equity firm (unlevered). Its value is $V_U = \$10{,}000{,}000$. The corporate tax rate is $T_c = 25\%$. The unlevered cost of equity is $K_{eu} = 12\%$. The firm decides to issue $\$3{,}000{,}000$ in perpetual debt at a cost of $K_d = 6\%$ and use the proceeds to repurchase equity.

Required:

1. Calculate the value of the levered firm (V_L) using M&M Proposition I with taxes.

$$V_L = V_U + T_c \times D = 10{,}000{,}000 + (0.25 \times 3{,}000{,}000) = \$10{,}750{,}000$$

2. Calculate the market value of equity (E) after the recapitalization.

$$E = V_L - D = 10{,}750{,}000 - 3{,}000{,}000 = \$7{,}750{,}000$$

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3- Calculate the debt-to-equity ratio (*D*/*E*).

D/E= debt/ Equity =3,000,000/7,750,000≈0.3871

4- Calculate the new cost of equity (*Ke*) using M&M Proposition II with taxes.

Ke=Keu+(1-Tc)×(Keu-Kd)×D/E

=12%+(1-0.25)×(12%-6%)×0.3871=12%+1.742%=13.742%

Example

An unlevered company has a cost of equity of 15%. It takes a loan at 10% and uses it to buy back equity, resulting in a D/E ratio of 1:1. The corporate tax rate is 20%. Calculate the new cost of equity and new WACC

- New cost of equity: $$K_e = K_{eu} + (1 - T_c) \times (K_{eu} - K_d) \times \frac{D}{E}$$
- Ke=15%+(1−0.20)×(15%−10%)×1=15%+4%=20%
- New WACC: $$\text{WACC} = \left(\frac{E}{V} \times K_e\right) + \left(\frac{D}{V} \times K_d \times (1 - T_c)\right)$$
- WACC=(0.5×20%)+(0.5×10%×(1−0.20))=10%+4%=14%
- The lower WACC increases firm value, illustrating the benefit of debt under a tax regime

B. Trade Off Theory

- **Formula (with distress costs):**
- $VL = VU + Tc \times D - PV \text{ of Financial Distress Costs}$
- **Implication:** Firms should use moderate levels of debt, not 100% debt, because excessive leverage increases bankruptcy risk

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ABC is currently unlevered. Its unlevered cost of equity is Keu=14%. It plans to issue debt at Kd=8%. The tax rate is *Tc*=30%. The firm's unlevered value is VU=$5,000,000. It will issue $1,000,000 of debt and use the proceeds to repurchase equity. After the recapitalization: Debt D=$1,000,000 & Equity E=$4,000,000

Required:

A. A. Calculate the debt-to-equity ratio D/E.

D/E=1,000,0004,000,000=0.25

B. Using M&M Proposition I without taxes, calculate *VL*.

VL=VU=$5,000,000

C. Using M&M Proposition II without taxes, calculate *Ke*.

Ke=Keu+(Keu−Kd)D/E= 14%+(14%−8%)(0.25)=15.5%

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D. Using M&M Proposition I with taxes, calculate *VL*.

VL=VU+TcD=5,000,000+0.30(1,000,000)=$5,300,000

E. Using M&M Proposition II with taxes, calculate *Ke*.

Ke=Keu+(1-Tc)(Keu-Kd)D/E=14%+(1-0.30)(14%-8%)(0.25)=14%+1.05%=15.05%

F. Using the Trade-off Theory, calculate *VL* if the present value of financial distress costs is $150,000

VL=VU+TcD-PV(Distress Costs)
=5,000,000+0.30(1,000,000)-150,000=$5,150,000

G. Calculate the WACC using the with-taxes cost of equity from part E

WACC: (EV)Ke+(DV)Kd(1-Tc) =(45)(15.05%)+(15)(8%)(1-0.30)=12.04%+1.12%=13.16%

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Approaches to capital structure

1] EBIT-EPS Approach

Sales revenues
(-) COGS
= Gross profit
(-) Operating Expenses
= Operating profit (EBIT)
(-)Interest
= Profit Before Tax
(-) Tax @ %
= Profit After Tax
(-)Preference Dividend
= Earnings to Equity
÷ Number of Shares
EPS

2] Share Value Approach

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1) EBIT-EPS Approach

Indifference Point Formula:

$$\frac{(\text{EBIT} - I_1)(1 - T)}{E_1} = \frac{(\text{EBIT} - I_2)(1 - T)}{E_2}$$

•**Example:** A company has:
Existing Capital Structure: Debt of ₹3,600,000 at 10% interest.
Expansion Requirement: ₹4,000,000 needed for expansion.
Tax Rate: 40%.
Current EBIT: ₹1,500,000.
Current Number of Shares: 800,000.
Required:
1] Apply alternative Financing Plans
Alternative I (Debt Financing): Issuing additional debt 12% interest
Alternative II (Preference Shares): Issuing preference shares with dividends 440,000 LE
Alternative III (Equity Shares): Issuing 250,000 new equity shares.
2] Calculation of Earnings Per Share (EPS)
3] Calculate EBIT according to Indifference point

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	Alternative I: Debt	
EBIT	₹1,500,000	Given
Interest (old + new)	(360,000 + 480,000) = (840,000)	(3,600,000* 10%) (4,000,000 * 12%)
Profit Before Tax	660,000	
Tax @ 40%	(264,000)	(660,000 * 40%)
Profit After Tax	396,000	
Preference Dividend	–	
Earnings to Equity	396,000	
Number of Shares	800,000	Given
EPS	**₹0.495**	

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	Alternative II: Preference Shares	
EBIT	₹1,500,000	
Interest (old + new)	(360,000)	(3,600,000*10%)
Profit Before Tax	1,140,000	
Tax @ 40%	(456,000)	(1,140,000*40%)
Profit After Tax	684,000	
Preference Dividend	(440,000)	Given
Earnings to Equity	244,000	
Number of Shares	800,000	
EPS	**₹0.305**	

	Alternative III: Equity Shares	
EBIT	₹1,500,000	Given
Interest (old + new)	(360,000)	(3,600,000*10%)
Profit Before Tax	1,140,000	
Tax @ 40%	(456,000)	(1,140,000*40%)
Profit After Tax	684,000	
Preference Dividend	–	
Earnings to Equity	684,000	
Number of Shares	1,050,000	800,000 + 250,000
EPS	**₹0.651**	

Indifference Point (Debt vs. Equity):

EBIT - 840,000/ 800,000 = EBIT - 360,000/ 1,050,000

EBIT ≈ ₹24,00,000, EPS under debt and equity plans are equal

- If expected EBIT > ₹24,00,000, the **debt plan** yields higher EPS (due to leverage).
- If expected EBIT < ₹24,00,000, the **equity plan** yields higher EPS.

Decision Rule:

- **Above the indifference point:** Use debt (financial leverage expands EPS).
- **Below the indifference point:** Use equity (avoids fixed interest burden).

2) Share Value Approach

$$P_0 = \frac{\text{EPS}}{r_s}$$

There is a 25% chance that Sales will total $400,000, a 50% chance that sales will total $600,000, and a 25% chance that sales will total $800,000. Fixed operating costs total $200,000, and variable operating costs equal 50% of sales. Total Assets $500,000. The Firm is assumed to have no current liabilities. The book Value per share of Common Stock is $20. The firm is considering seven alternative capital structures. The debt ratios associated are at 0%, 10%, 20%, 30%, 40%, 50%, and 60% levels.

Choose the optimal capital structure based on the EPS Approach and the Share Value
Approach (tax=40%)

Capital structure debt ratio	Interest rate on *all* debt (2)
0%	0.0%
10	9.0
20	9.5
30	10.0
40	11.0
50	13.5
60	16.5

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Step 1: Calculate the amount of debt & equity, No. of shares associated with each capital structure

	Capital structure ($000)			Shares of common stock outstanding (000)
Debt ratio	Total assets[a]	Debt [(1) × (2)]	Equity [(2) − (3)]	[(4) ÷ $20][b]
(1)	(2)	(3)	(4)	(5)
0%	$500	$ 0	$500	25.00
10	500	50	450	22.50
20	500	100	400	20.00
30	500	150	350	17.50
40	500	200	300	15.00
50	500	250	250	12.50
60	500	300	200	10.00

[a]Because the firm, for convenience, is assumed to have no current liabilities, its total assets equal its total capital of $500,000.

[b]The $20 value represents the book value per share of common stock equity noted earlier.

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Step 2: Calculate annual interest payments associated with alternative capital structure

Capital structure debt ratio	Debt ($000) (1)	Interest rate on *all* debt (2)	Interest ($000) [(1) × (2)] (3)
0%	$ 0	0.0%	$ 0.00
10	50	9.0	4.50
20	100	9.5	9.50
30	150	10.0	15.00
40	200	11.0	22.00
50	250	13.5	33.75
60	300	16.5	49.50

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Step 3: Calculate sales and associated EBIT

Probability of sales	0.25	0.50	0.25
Sales revenue	$400	$600	$800
Less: Fixed operating costs	200	200	200
Less: Variable operating costs (50% of sales)	200	300	400
Earnings before interest and taxes (EBIT)	$ 0	$100	$200

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Step 4: Calculate EPS for different debt ratios
Debt ratio= 0%

Probability of EBIT		0.25	0.50	0.25
Debt ratio = 0%				
EBIT (Table 13.9)		$ 0.00	$100.00	$200.00
Less: Interest (Table 13.11)		0.00	0.00	.00
Net profits before taxes		$ 0.00	$100.00	$200.00
Less: Taxes (T = 0.40)		0.00	40.00	80.00
Net profits after taxes		$ 0.00	$ 60.00	$120.00
EPS (25.0 shares, Table 13.10)		$ 0.00	$ 2.40	$ 4.80
Expected EPS[a]	$2.40			

Expected EPS = Sum (EPS*Probability)

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Step 4: Calculate EPS for different debt ratios
Debt ratio= 30%

Probability of EBIT		0.25	0.50	0.25
Debt ratio = 30%				
EBIT (Table 13.9)		$ 0.00	$100.00	$200.00
Less: Interest (Table 13.11)		15.00	15.00	15.00
Net profits before taxes		($15.00)	$ 85.00	$185.00
Less: Taxes (T = 0.40)		(6.00)[b]	34.00	74.00
Net profits after taxes		($ 9.00)	$ 51.00	$111.00
EPS (17.50 shares, Table 13.10)		(0.51)	$ 2.91	$ 6.34
Expected EPS[a]	$2.91			

Expected EPS = Sum (EPS*Probability)

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Step 4: Calculate EPS for different debt ratios
Debt ratio= 60%

Probability of EBIT		0.25	0.50	0.25
Debt ratio = 60%				
EBIT (Table 13.9)		$ 0.00	$100.00	$200.00
Less: Interest (Table 13.11)		49.50	49.50	49.50
Net profits before taxes		($49.50)	$ 50.50	$150.50
Less: Taxes (T = 0.40)		(19.80)[b]	0.20	60.20
Net profits after taxes		($29.70)	$ 30.30	$ 90.30
EPS (10.00 shares, Table 13.10)		($ 2.97)	$ 3.03	$ 9.03
Expected EPS[a]	$3.03			

[b]It is assumed that the firm receives the tax benefit from its loss in the current period as a result of applying the *tax loss carryback* procedures specified in the tax law but not discussed in this text.

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Step 5: Calculate Share Value (EPS / r) for different Debt Ratios.

Capital structure debt ratio	Expected EPS (from column 1 of Table 13.13) (1)	Estimated required return, r_s (from column 2 of Table 13.14) (2)	Estimated share value [(1) ÷ (2)] (3)
0%	$2.40	0.115	$20.87
10	2.55	0.117	21.79
20	2.72	0.121	22.48
30	2.91	0.125	23.28
40	3.12	0.140	22.29
50	3.18	0.165	19.27
60	3.03	0.190	15.95

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Conclusion

Based on EPS-Approach:

EPS is maximized at Debt ratio 50%. Hence, the optimal capital structure is 50% Debt and 50% Equity.

Based on Share-Value Approach:

The maximum share value occurs at the capital structure associated with a debt ratio of 30%. Hence, the optimal capital structure is 30% debt and 70% Equity

Thank You….
See you next class